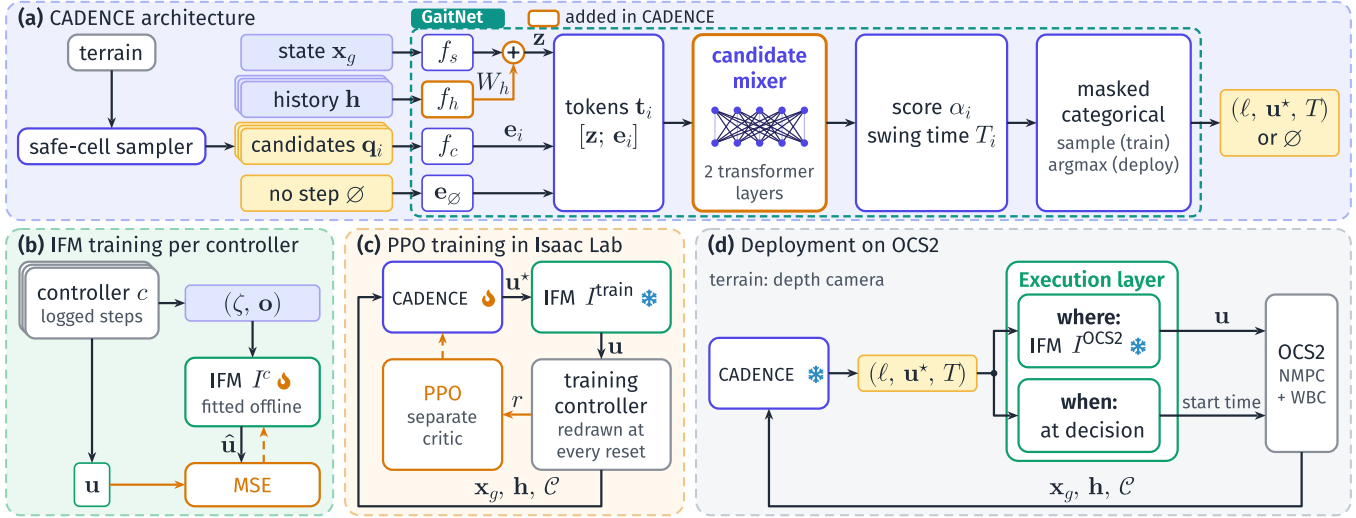

(a) CADENCE architecture
GaitNet
added in CADENCE
terrain
safe-cell sampler
state $\mathbf{x}_g$
history $\mathbf{h}$
candidates $\mathbf{q}_i$
no step $\varnothing$
f_s
f_h
f_c
$\mathbf{e}_\varnothing$
W_h
$\mathbf{z}$
$\mathbf{e}_i$
tokens $\mathbf{t}_i$
$[\mathbf{z}; \mathbf{e}_i]$
candidate mixer
2 transformer layers
score α_i
swing time T_i
masked categorical
sample (train)
argmax (deploy)
$(\ell, \mathbf{u}^\star, T)$ or $\varnothing$
(b) IFM training per controller
controller c
logged steps
$(\zeta, \mathbf{o})$
IFM I^c
fitted offline
$\hat{\mathbf{u}}$
$\mathbf{u}$
MSE
(c) PPO training in Isaac Lab
CADENCE
$\mathbf{u}^\star$
IFM I^{train}
$\mathbf{u}$
PPO
separate critic
r
training controller
redrawn at every reset
$\mathbf{x}_g$, $\mathbf{h}$, $\mathcal{C}$
(d) Deployment on OCS2
terrain: depth camera
CADENCE
$(\ell, \mathbf{u}^\star, T)$
Execution layer
where:
IFM I^{OCS2}
when:
at decision
$\mathbf{u}$
start time
OCS2
NMPC
+ WBC
$\mathbf{x}_g$, $\mathbf{h}$, $\mathcal{C}$
data flow
training signal
parameter update
trained
frozen